

Taresh Guleria

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PROFESSIONAL SUMMARY

Process engineer and Ph.D. candidate in Industrial Engineering at Texas A&M (Dec 2026), specializing in metal additive manufacturing: laser powder bed fusion (LPBF) of NiTi shape memory alloys (nitinol), NiTiHf, and Ti-6Al-4V. Reduced qualification cycles ~80% and process waste ~75% via DOE and closed-loop control while maintaining >99% specification compliance. [O-1 & EB-2 NIW eligible]

EXPERIENCE

Graduate Research Assistant (Ph.D.)

Jan 2023 – Present

Texas A&M University, Dept. of Industrial & Systems Engineering

College Station, TX

- Cut qualification coupons 80% by establishing defect-free LPBF process windows for 2 alloy systems (NiTi, NiTiHf) with DOE and SPC, removing rework through Lean/Six Sigma methods.
- Achieved first-pass qualification on a Los Alamos National Laboratory containment project with functionally graded NiTi structures carrying spatially controlled transformation temperatures and modulus gradients for sealing and damping.
- Raised build yield ~60% with in-situ thermal monitoring and closed-loop melt pool control for real-time defect detection; protocols now scaled to production builds.
- Cut field qualification time 40% for new material submissions with SOPs and robustness protocols documented in ISO/ASTM AM standards contributions.
- Ran process development with 3 national labs (ORNL, LANL, Sandia) and an NSF program; 7 peer-reviewed papers, 2 first-author.

R&D Engineering Intern, Metals R&D & Government Projects

Jan 2026 – May 2026

EOS of North America

Pflugerville, TX

- Removed the re-qualification bottleneck on the DELTA-QUAL and ACCELERATE (America Makes) Ti-6Al-4V programs, cutting coupon testing 80% and raising throughput 65% with specs held, using in-situ monitoring feedback and DOE parameter sets.
- Started VITAL (LIFT program), a closed-loop process control and property-prediction framework aligning qualification pathways and tool efficiency metrics across competing OEM platforms.
- Cut virgin feedstock use ~75% on RECAP (America Makes) through powder reuse, LPBF parameter optimization, and powder composition tracking, while holding tensile, fatigue, and cyclic reliability specs.

Research Intern (NSF INTERN Fellow), Manufacturing Demonstration Facility

May 2024 – Aug 2024

Oak Ridge National Laboratory

Knoxville, TN

- Reached >99% density in LPBF NiTi by optimizing parameters across 100+ samples, and established process-microstructure-property relations governing cyclic reliability under multiaxial loading.
- Traced defect formation and strain localization to root causes using EBSD, in-situ XRD, and DIC full-field strain, enabling predictive yield and fatigue-life models for production validation.
- Turned the summer fellowship into 1 peer-reviewed paper (Additive Manufacturing Letters, 2025) and 1 conference presentation (Solid Freeform Fabrication Symposium, 2025); microstructure-control strategies fed MDF's roadmap for scaling NiTi LPBF.

EDUCATION

Ph.D., Industrial Engineering, Texas A&M University, College Station, TX, GPA 4.0/4.0

Expected Dec 2026

M.S., Industrial and Systems Engineering, Texas A&M University, College Station, TX, GPA 3.45/4.0

Dec 2024

TECHNICAL SKILLS

Additive Manufacturing: LPBF (EOS M290/M400, AMCM M290-2, 3D Systems ProX 200), EOS OT and Smart Fusion monitoring, nitinol (NiTi/NiTiHf), Ti-6Al-4V, functionally graded materials, powder reuse

Process Development & Manufacturing: Design of experiments (DOE), process window qualification, statistical process control (SPC), in-situ and in-line monitoring, closed-loop adaptive control, defect root-cause analysis, yield improvement, production readiness and process robustness validation

Metrology & Characterization: SEM, EBSD, in-situ XRD, DSC, DIC full-field strain, FEA thermomechanical, fatigue and cyclic testing, mechanical property prediction

Software & Tools: EOSPRINT, Dyndrite, 3DXpert, Siemens NX, ANSYS Additive Suite, CREO, AVEVA-PDMS, Python, MATLAB

SELECTED PUBLICATIONS & RECOGNITION

Publications: [Google Scholar](#): seven peer-reviewed articles, including *Additive Manufacturing* (2026, first author), *Acta Materialia* (2025), *Materials Today* (2026)

Recognition: 2nd of 80+, ASTM AM Poster Competition 2025 | NSF INTERN Fellowship, ORNL 2024 | Best Research Presentation, AM Symposium 2024